

Radio Tracker Journal 1

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Abstract

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By using capacitors known charge rates as timers, the device can measure distance accurately from a central hub to an emitter that will bounce the signal back. If the hub has 3 receivers, we can triangulate the position of the emitter.

1 Introduction

Using radio waves to triangulate position has been a well developed practice, however all methods are clock-speed dependent, or the 3 receivers are miles apart. The purpose of this project is to allow an analog signal to time a signal's output and return. When the hub sends out the signal, the capacitor will start to receive charge from a stable, known power source. Then as the signal returns to the hub, the charging will be cut off and the capacitor will hold the value until the ADC can read the value. Of course there will be leakage, but if the ADC can attain the value fast enough, and with enough pulses, the device will more accurately determine the position of the emitter.

2 Subsystems

2.1 Main Hub

The main hub will need 1 radio generator, and 3 radio receivers. Each receiver will be in an equilateral triangle formation around the 1 generator. Each receiver will have a capacitor timer at full discharge at the start. The responsive signal from the emitter will be sent to the main controller as a flag to measure the voltages of each capacitor and begin calculations. After calculations are completed, the CT is shorted and the cycle starts again.

2.2 Capacitor Timer (CT)

At the same moment that the hub's radio generator outputs a signal, the capacitors will start charging. This charging will not stop until a response is received from the emitter, in which the flow of electricity will be cut off and the capacitor will be floating.

2.3 Emitter

This emitter will be a passive device that waits for the signal from the main hub. Once received it automatically reply's with a separate frequency towards the main hub.

3 Timing calculations

The time it takes to charge the capacitor can be sectioned by "time constants" RC . With a higher RC , the capacitor will take longer to charge. The main problem with the circuit will be seeing differences in the charges. This difference will give us a "resolution" for how accurate the tracker can be. The 3 main factors of the resolution is the RC time frame, the distance between each receiver, and the resolution of the ADC that will take in the measurement. Since the radio wave will be traveling at the speed of light, this time window is very short. Therefore, We also need a quick input current to charge the capacitor fast enough to make a difference during the small windows in which one receiver has obtained the signal, and another has not.

It is important to note that a capacitors charge is logarithmic. Our optimal point of noting the difference between receivers is before the first time constant. In this range, the voltage changes drastically and will be easier to measure differences when they occur.

3.1 Calculating Max Δt between receivers based on distance from center of Hub.

Since it will be placed in a equilateral triangle, the distance between 2 receivers will be $l * \sqrt{3}$. Therefore with a equal length of 15 cm: the distance between will be 25.98 cm.

$$d = 15cm * \sqrt{3} = 25.98cm$$
$$\Delta t = \frac{d}{c} = \frac{25.98cm}{3 * 10^{11}cm/s} = 86.6ps$$

This is a very short amount of time, something that no computer can keep up with. However, we want the range of time measure to be within $t = 0 \rightarrow t = RC$. In this range, the ΔV will give us a higher resolution of data. The problem then becomes a "bottleneck" of timing. Assuming the best situation where the radio wave has to propagate the longest distance (giving us the clearest difference in capacitor charges), there is no transistor that can open or close the capacitors charge with minimal noise. Which brings us the question, how will the distance be calculated?

4 Calculating Distance

Once the values are in the computer, we can revert it back to time based off the voltage values. Then we may subtract known delays, such the time it took for a signal to be amplified or a switch to turn on. However, there still is the issue of switch inconsistency and noises. Luckily, if we take enough samples, we can take the average, thus eliminating any low level variance.

$$t_{total} = t_{travel} + t_{Delay} + t_{noise}$$

For calculation, we take the average of all the values for the distance (after subtracting known delays). For direction, we take each receivers value, and subtract it from the average, and use those difference in values to evaluate from which direction it came from. Therefore:

$$distance = \sum_{i=1}^3 \frac{t_i - t_{delay}}{3}$$

5 Direction Estimation Using Equilateral Geometry

Consider three known reference points forming an equilateral triangle of side length α . Without loss of generality, the coordinates of the reference points are chosen as

$$P_1 = (0, 0), \quad P_2 = (\alpha, 0), \quad P_3 = \left(\frac{\alpha}{2}, \frac{\sqrt{3}}{2}\alpha\right). \quad (1)$$

Let an unknown point $P = (x, y)$ be located at distances d_1 , d_2 , and d_3 from P_1 , P_2 , and P_3 , respectively. Rather than estimating the exact coordinates of P , only the *direction* of P relative to the triangle is desired.

5.1 Centroid Reference

The centroid of the equilateral triangle is given by

$$C = \left(\frac{\alpha}{2}, \frac{\sqrt{3}}{6}\alpha\right). \quad (2)$$

The direction of the unknown point is defined as the angle θ of the vector from the centroid C toward P .

5.2 Distance Normalization

To remove absolute range dependence, the mean distance is computed as

$$\bar{d} = \frac{d_1 + d_2 + d_3}{3}. \quad (3)$$

Relative distance deviations are then defined as

$$\Delta_i = d_i - \bar{d}, \quad i = 1, 2, 3. \quad (4)$$

These deviations encode directional information while rejecting common-mode distance bias.

5.3 Directional Vector Construction

Define unit vectors from the centroid toward each vertex:

$$\hat{u}_1 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad \hat{u}_2 = \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix}, \quad \hat{u}_3 = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}. \quad (5)$$

A direction vector pointing toward the unknown point is constructed as

$$\vec{v} = \Delta_1 \hat{u}_1 + \Delta_2 \hat{u}_2 + \Delta_3 \hat{u}_3. \quad (6)$$

This vector is proportional to the direction of P and does not depend on absolute range.

5.4 Direction Angle

The bearing angle of the unknown point relative to the centroid is obtained as

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right) \quad (7)$$

where v_x and v_y are the Cartesian components of $\vec{v}$.

5.5 Discussion

This approach exploits the symmetry of the equilateral triangle to extract directional information using only relative distance differences. It avoids explicit coordinate estimation, reduces sensitivity to uniform measurement bias, and is well suited for embedded and RF localization systems.

6 The Resolution VS Range Problem

This does bring a major concern, that a capacitor can have either high resolution of time, or a large capacity of time. While the travel time will be in range of 100's ps, the delays found in amplifiers will be in 100's ns. We want the capacitors to have a resolution of ps since the wave travels so fast.

6.1 Solution A

Instead of starting the CT at the same time as the waves start, we delay the start of the CT the same amount we expect the devices in the system will cause. Therefore, the CT will only be tracking $t_{travel} + t_{noise}$. However we can only do this so much since we still need ns of time for a transistor to start and stop the CT. A speed that can only be done in ns. Therefore, we need to increase capacity.

6.2 Solution B

We can add more capacitors in parallel to each other, with diodes in between them. The diodes will require a certain voltage threshold before it allows current to pass, and we can add up the timers in sequential order. I was going to use transistors to manipulate capacitors in a "binary counter" fashion, however, each transistor will cause ns of delay. With diodes, we can make it a linear counter, with at best 50 ps of delay with each diode. This creates a hard wall of resolution in the 50 ps range. This can be fine however, with a slightly increased distance between each receiver.

There is a problem however. The diode solution is not strictly linear after we do more than 1 stage of it. The first stage will work fine, we wait till it hits its desired threshold of ΔV , then it starts charging the next parallel CT. However, as the 2nd CT starts its

charge, it will eventually cause the diode to close up since the ΔV will approach 0. It becomes more and more problematic with the more sections we add. It can be said that these conditions and charge values can be analyzed, and 1, 0, 1, 0 thus we can still use it, get a number of CT voltage readings and reverse analyze it to see how much time it took to get to that stage. What matters here is consistency that we can simulate, and thus find the time it would have taken to attain those said voltages. It could be said that we can use a neural net machine learning model trained off of simulations to guess the timings based off what the simulations said. I would prefer hard calculations but this is the only way to evaluate a high capacity, high resolution solution with what we have.

6.3 Solution C

The answer could be in the problem itself. We will experience known delays when using switches and amplifiers. What if, we used option A, but instead of having buffers before we being the CT, the delays are in the CT itself.

For example, if we implement a binary counter solution, where we use transistors to change the direction of charge and discharge certain elements, that delay that caused by the switch turning on and off can be known when we evaluate the charge. Therefore, if we get values of .5, 0, .9, 0, we know that we are in between the binary count of 0, 0, 1, 0 and 1, 0, 1, 0, which would have taken 4 known switches + what ever time it took for use to get to 0.5 charge of the first CT. We can evaluate the time to be roughly:

$$t = n * t_{delays} + t_{currentCT} + b_{CTChargetimes}$$

Where n = number of delays the transistors took, b is lower end of the binary number we are on, and current CT is the capacitors value of what it seems to be the current capacitor charging.

There are of course, a number of issues with t 1, 0, 1, 0his, what happens if we stop the timer, mid way of a transistor switching? This is a massive time frame considering it can take nanoseconds. There might be leakage into a CT we didnt intend for, which can cause false results, however this does provide a solution that dramatically increases our capacity without really diminishing.

However I will admit, bottom line, the transistors simply might not be fast enough. The time it takes a transistor to close would ruin the ps of resolution we get from the capacitors. And with the binary solution, we are simply adding a digital counter to the problem which is certainly not fast enough. Its already been attempted.

6.4 Decision

Out of all my options and ideas, i believe the diode solution is the best. The delay it causes to allow current to flow is the fastest solution. Therefore it will ruin the resolution the least. The behavior can be simulated and predicted which does make it a good route to evaluate the time. Based off the behavior of the system, I might use a algorithm to evaluate which simulated time matches the results best, or a AI model to predict it. Either option can be a POST calculation since all we need is the data first. We can then average it to rid of the noise.

I will then combine the solutions of option A and B to optimize the resolution and capacity of the system.

I will design and simulate the ideas presented here and bring a update in journal 2.